

Data Modeling

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Dept: Data Engineering

- Adding necessary tables from dataset

The screenshot displays the Microsoft Power BI Desktop interface. The main view shows a data table with the following columns: **CustomerKey**, **Prefix**, **FirstName**, **LastName**, **BirthDate**, and **MaritalStatus**. The table contains 24 rows of data. The left-hand pane shows the 'Queries' list with 'Customers' selected. The right-hand pane shows the 'Query Settings' for the 'Customers' query, with the 'APPLIED STEPS' list containing 'Removed Duplicates'.

	CustomerKey	Prefix	FirstName	LastName	BirthDate	MaritalStatus
1	11000	MR.	JON	YANG	08-04-1966	M
2	11001	MR.	EUGENE	HUANG	14-05-1965	S
3	11002	MR.	RUBEN	TORRES	12-08-1965	M
4	11003	MS.	CHRISTY	ZHU	15-02-1968	S
5	11004	MRS.	ELIZABETH	JOHNSON	08-08-1968	S
6	11005	MR.	JULIO	RUIZ	05-08-1965	S
7	11007	MR.	MARCO	MEHTA	09-05-1964	M
8	11008	MRS.	ROBIN	VERHOFF	07-07-1964	S
9	11009	MR.	SHANNON	CARLSON	01-04-1964	S
10	11010	MS.	JACQUELYN	SUAREZ	06-02-1964	S
11	11011	MR.	CURTIS	LU	04-11-1963	M
12	11012	MRS.	LAUREN	WALKER	18-01-1968	M
13	11013	MR.	IAN	JENKINS	06-08-1968	M
14	11014	MRS.	SYDNEY	BENNETT	09-05-1968	S
15	11015	MS.	CHLOE	YOUNG	27-02-1979	S
16	11016	MR.	WYATT	HILL	28-04-1979	M
17	11017	MRS.	SHANNON	WANG	26-06-1944	S
18	11018	MR.	CLARENCE	RAI	09-10-1944	S
19	11019	MR.	LUKE	LAL	07-03-1978	S
20	11020	MR.	JORDAN	KING	20-09-1978	S
21	11021	MS.	DESTINY	WILSON	03-09-1978	S
22	11022	MR.	ETHAN	ZHANG	12-10-1978	M
23	11023	MR.	SETH	EDWARDS	11-10-1978	M
24	11024	MR.	BLISS	VIE	17-02-1979	M

13 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 11:11

Customers to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

CustomerKey	OrderDate	OrderLineItem	OrderNumber	OrderQuantity	ProductKey	Sales_year
13622	2021-01-03	1	SO48819	1	352	2021
13960	2021-01-13	1	SO48901	1	352	2021
13953	2021-01-14	1	SO48909	1	352	2021

To table

Customers

AnnualIncome	BirthDate	CustomerKey	EducationLevel	EmailAddress	FirstName	Gender
40000	22 April 1973	11479	Bachelors	darryl6@adve...	DARRYL	M
40000	22 May 1972	11483	Bachelors	calvin13@adv...	CALVIN	M
40000	22 November...	11584	Bachelors	devin33@adv...	DEVIN	M

Cardinality Many to one (*:1)

Cross-filter direction Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

Save Cancel

Products to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

CustomerKey	OrderDate	OrderLineItem	OrderNumber	OrderQuantity	ProductKey	Sales_year
13622	2021-01-03	1	SO48819	1	352	2021
13960	2021-01-13	1	SO48901	1	352	2021
13953	2021-01-14	1	SO48909	1	352	2021

To table

Products

ModelName	ProductColor	ProductCost	ProductDescri...	ProductKey	ProductName	ProductPr
Sport-100	Red	13.0863	Universal fit, ...	214	Sport-100 Hel...	34.99
Sport-100	Black	12.0278	Universal fit, ...	215	Sport-100 Hel...	33.6442
Sport-100	Blue	12.0278	Universal fit, ...	220	Sport-100 Hel...	33.6442

Cardinality Many to one (*:1)

Cross-filter direction Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

Save Cancel

Territories to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

OrderItem	OrderNumber	OrderQuantity	ProductKey	Sales_year	StockDate	TerritoryKey
	SO48819	1	352	2021	2020-09-06	9
	SO48901	1	352	2021	2020-09-19	9
	SO48909	1	352	2021	2020-09-22	9

To table

Territories

Continent	Country	Region	SalesTerritor...
North America	United States	Northwest	1
North America	United States	Northeast	2
North America	United States	Central	3

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

Products to Sales (1 to *)

Edit relationship

Select tables and columns that are related.

From table

Sales Data

CustomerKey	OrderDate	OrderLineItem	OrderNumber	OrderQuantity	ProductKey	Sales_year
13622	2021-01-03	1	SO48819	1	352	2021
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Sport-100	Black	12.0278	Universal fit, ...	215	Sport-100 Hel...	33.6442
Sport-100	Blue	12.0278	Universal fit, ...	220	Sport-100 Hel...	33.6442

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

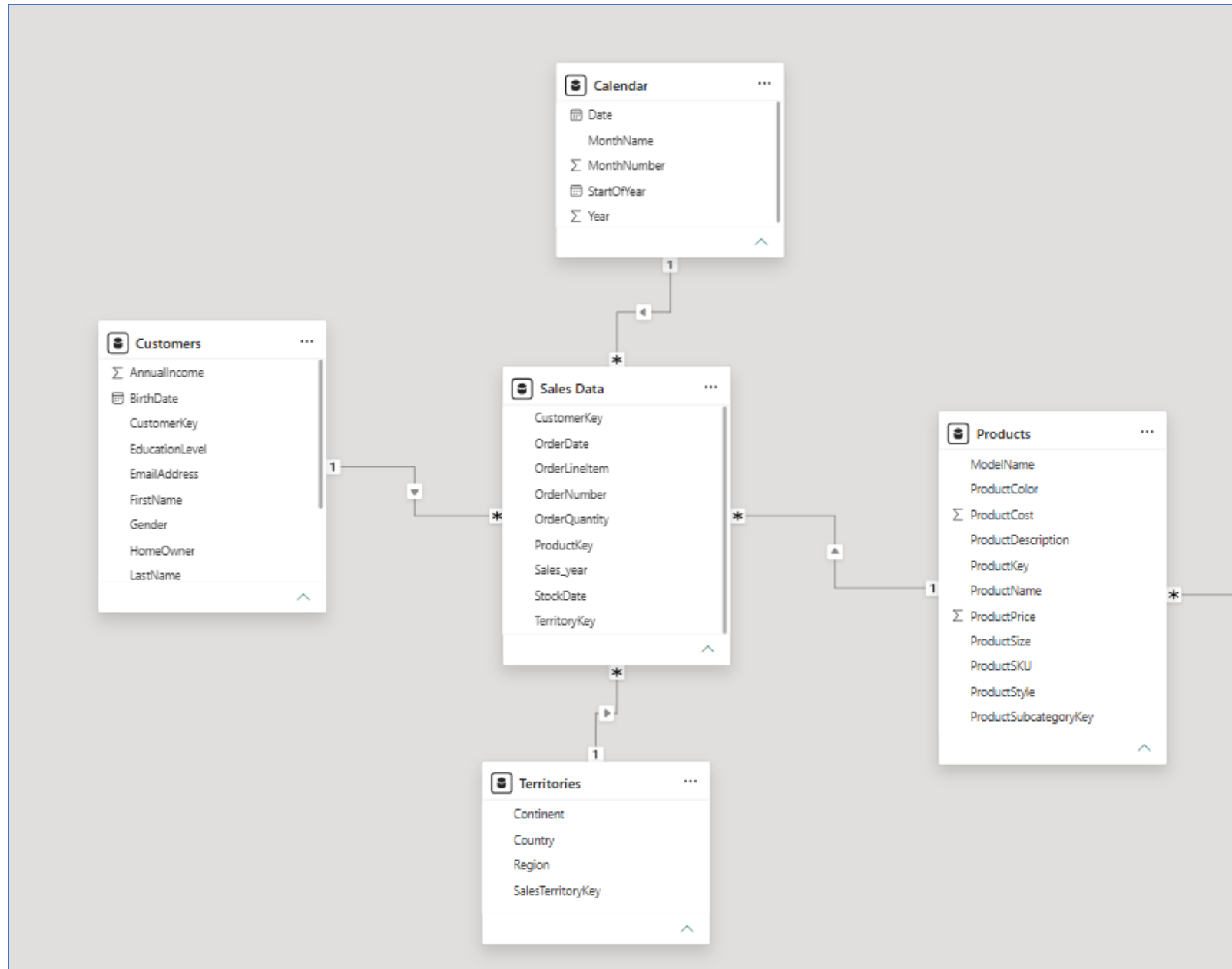
☐ Apply security filter in both directions

☐ Assume referential integrity

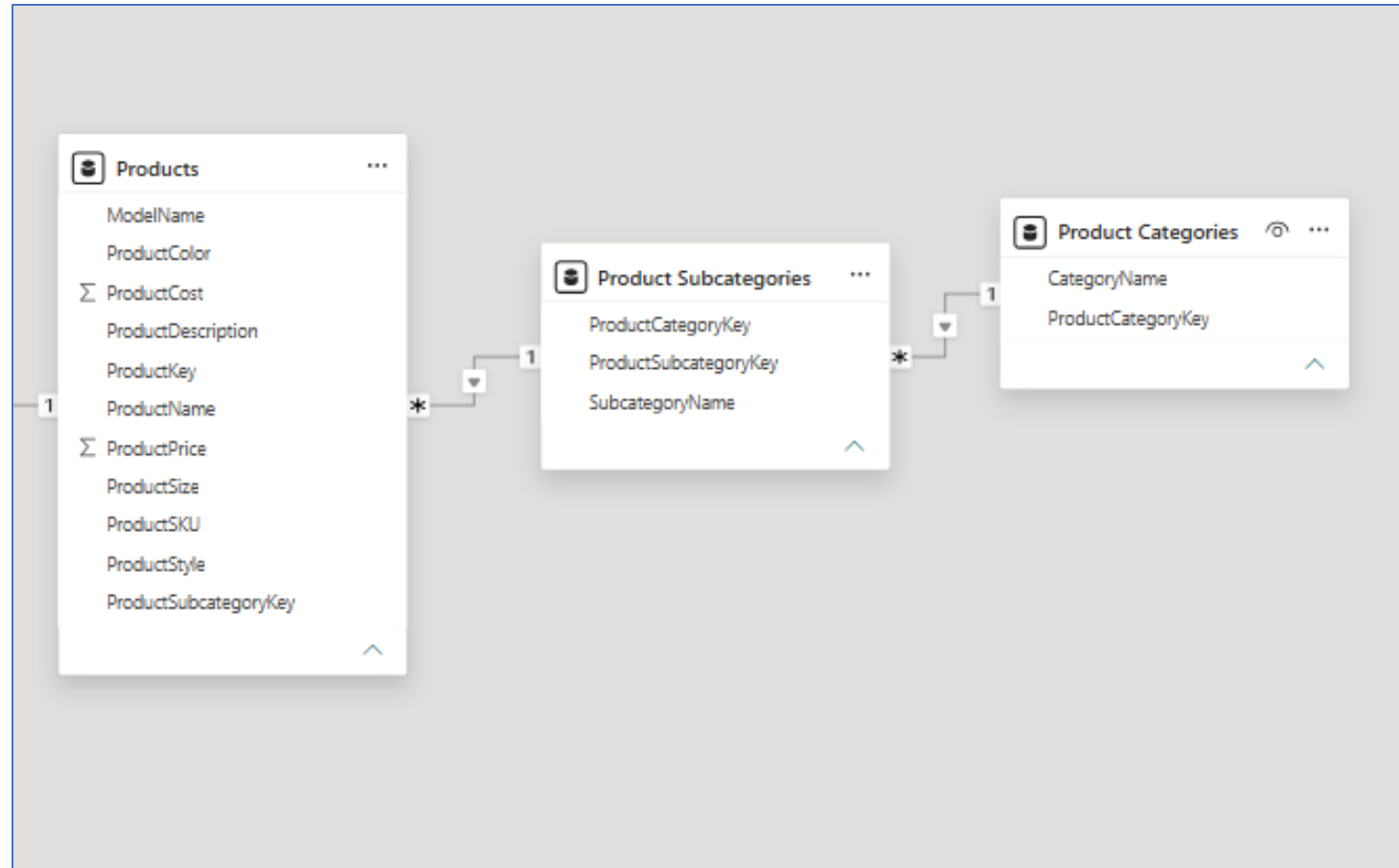
Save

Cancel

Star Schema



Products – subcategories – categories



Adding necessary columns in Calender table for hierarchy

Custom Column

Add a column that is computed from the other columns.

New column name

StartOfMonth

Custom column formula ⓘ

= DateDate.StartOfMonth([Date])

Available columns

Date

MonthName

MonthNumber

StartOfYear

Year

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Custom Column

Add a column that is computed from the other columns.

New column name

StartOfWeek

Custom column formula ⓘ

= Date.StartOfWeek([Date], Day.Monday)

Available columns

Date

MonthName

MonthNumber

StartOfYear

Year

StartOfMonth

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

Creating hierarchy from startOfYear

